

CENTRAL UNIVERSITY

FAITH • INTEGRITY • EXCELLENCE

SCHOOL OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY

DEVELOPMENT OF A SAAS-BASED LEARNING MANAGEMENT SYSTEM (LMS) FOR UNIVERSITY STUDENTS

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DECLARATION

Students Declaration

I declare that this project work is because of our original research and that none of it has been presented for another degree in this University or elsewhere. We are responsible for any errors and omissions detected.

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Supervisor's Declaration

I, at this moment, declare that the preparation of the project work was supervised by the guidelines of supervision of project work laid by Central University

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Chapter One

INTRODUCTION

1.1 Background of the Study

In today's era of digitalization, education has incurably altered how institutions visualize teaching and learning. Learning management system (LMS) have evolved from static repository of course material to dynamic ecosystems designed to foster collaboration, accessibility and engagement. however, as higher education combat with the expectation of generation z and alpha learners, digital native raised in era of Instagram, TikTok and messaging, traditional learners platform appears outdated. These systems, typically built on nice models struggle for fluid and community nature of modern social media. (Dabbagh & Kitsantas, 2012; Veletsianos, 2020). The limitations of conventional LMS are well-explained. A study by Siemens (2013) and Bozkurt et al. (2023) point to their inflexible structure, which hinder organic peer-to-peer interaction and reduces learning to transactional processes (uploading assignment and accessing lectures). For instance, forum discussion platforms like Moodle or Blackboard tend to resemble bureaucratic workflows rather than natural conversation, often resulting student in disengagement. A 2023 EDUCUASE report revealed that 72% of student find traditional LMS interface detaching compared to the user-friendly design of social apps like Discord or Slack. This discontent reflect a significant gap between institutional tools and modern student expectations. To enter Software-as-a-Service (SaaS) models by leveraging cloud computing, SaaS-based LMS solutions eliminate the financial and technical barriers of legacy systems, offering institutions scalable, subscription based platforms with automatic updates and global accessibility (Alshahrani & Walker, 2022). Yet, while SaaS addresses infrastructural inefficiencies, it does not inherently resolve educational limitations. This gap has spurred interest in social media-inspired LMS design a paradigm shift towards platforms that blend academic difficulties with the interactivity of platforms like Facebook and LinkedIn. A study by Li et al. (2022) demonstrates that LMS platforms integrating features such as activity feeds, peer endorsement(likes), and real-time chat see 45% increase in student participation and a 30% improvement in collaborative outcomes. This study and the system being developed aim to address three key challenges in modern education:

- Engagement Crisis, Passive, one-size-fit-all LMS designs fail to motivate learners accustomed to personalized, interactive digital experiences (Pappas, 2021).

- Educational Stagnation, Traditional LMS platforms after reinforce instructor-centered models, neglecting social constructivist principle that emphasize peer-driven knowledge creation (Vygotsky, 2018; Dron & Anderson, 2014).

- Technological Inflexibility, Legacy system lack the agility to support emergent tools like AI-driven analytics or real-time collaboration, limiting their adaptability to future trends (EDUCAUSE 2023).

The LMS being developed in this project addresses these issues by embedding social media mechanics into its core architecture. Key innovations include:

- Activity Feeds - centralized stream for course updates, peer posts, and multimedia content, mimicking the "timeline" structure of platforms like Twitter.
- Reaction-Based Feedback - Students and instructors can "like," "celebrate," or "question" posts using emojis, fostering micro-interactions that mirror informal learning communities.
- Real-Time Messaging -Integrated chat and video conferencing tools enable spontaneous collaboration, reducing reliance on external apps like WhatsApp or Zoom.
- User-Generated Content - Learners can create and share posts, info-graphics, or video reflections, democratizing content creation and aligning with connectivist pedagogy (Siemens, 2005).

The system is built on a Javascript framework The frontend employs Next js to create a responsive, app-like interface, while supabase handles real-time database operations for messaging and notifications. Clerk and external library for User Authentication Crucially, the design is grounded in self-determination theory which posits that intrinsic motivation thrives when users feel autonomous, competent, and socially connected. For example, "likes" and comments act as lightweight social validation, while peer messaging builds camaraderie and reduces the isolation often reported in online learning

1.2 Problem Statement

Currently, despite the widespread adoption of Learning Management Systems (LMS) in educational sectors, contemporary platforms struggle to meet the evolving demand of today's digital native student. Three interrelated challenges persist, undermining the potential of online learning environment Traditional LMS platforms, designed around stratified content delivery like lecture uploads and assignment drop-box, fail to replicate the interactive, participatory experiences student encounter on social media. A study by Pappas (2021) reveals that 62% of university students perceive conventional LMS interfaces as passive and monotonous, leading to disengagement. For instance, discussion forum in systems like Moodle or Blackboard often suffer from low participation rates, with only 15%-20% of students actively contributing to threaded discussions (Bower et al. 2017). This passivity contradicts the educational frameworks like social constructivism (Vygotsky,2018), which emphasize active knowledge co-creation through dialogue. Existing LMS platforms prioritize instructor-to-student communication, neglecting peer-to-peer and group-based collaboration. A 2023 EDUCAUSE report found that 78% of students rely on external tools like WhatsApp, Discord, or Google Docs to collaborate, fragmenting the learning experience. This disconnect reflects a structural flaw in LMS design while platforms like Canvas offer basic group features, they lack real-time, informal interaction tools (e.g., instant messaging, reaction-based feedback) that foster spontaneous community-building (Dron & Anderson, 2014). Such limitations contradict connectivist

principles (Siemens, 2005), which position learning as a networked, socially embedded process. Most LMS platforms employ a one-size-fits-all approach, offering limited adaptability to diverse learning styles, discipline or student demographics. For example, STEM courses requiring real-time coding collaboration or art classes emphasizing multimedia portfolios cannot easily tailor rigid LMS templates to their needs. A study by Li et al.(2022) demonstrate that personalized learning pathways improves retention rate by up to 30%, yet traditional systems lack AI-driven analytic or modular architectures to support such customization The consequences of failing to address these challenges have significant academic and institutional impacts. Institutions relying on static learning management system (LMS) experience 20-25% higher dropout rates in online courses compared to those using interactive and community-driven platforms (Jordan,2015). Additionally, student satisfaction declines in traditional LMS environment, with 68% of learners reporting feeling isolated due to lack of informal social spaces similar to campus common areas (Gunawardena et al.2009). Furthermore, inflexible LMS designs disproportionately disadvantage non-traditional students, such as working adults and rural learners, who require mobile-first interfaces and asynchronous collaboration tools to engage effectively (Warschauer, 2004). Addressing these limitations is crucial to improving retention, satisfaction, and equitable access to education.

The Gap in Current Solutions

While SaaS-based LMS platforms (e.g., Canvas Instructure, D2L) have improved scalability and cost-efficiency, they often replicate the pedagogical limitations of legacy systems. For example, despite cloud adoption, 89% of SaaS LMS users report dissatisfaction with “static” discussion boards and “impersonal” feedback mechanisms (EDUCAUSE, 2023).

1.3.1 Main Objective

The primary objective of this project is to design and implement a socially driven Learning Management System (LMS) that bridges the gap between formal education and the interactive, community-oriented digital environments students inhabit daily. By integrating features inspired by social media platforms such as real-time messaging, collaborative activity feeds, and peer-to-peer interaction tools

1.3.2 Specific Objectives

1. To Provide educators with intuitive tools to create, organize, and manage course materials efficiently, allowing them to focus on delivering high-quality content.
2. To Implement a suite of social features, including posts, comments, likes, and real-time chat, to encourage collaboration and build a community around learning
3. To Offer white-label capabilities, allowing groups to establish unique, branded learning spaces within the platform

1.4 Research Question

How does integrating social media features into an LMS affect student engagement compared to traditional LMS platforms?

What impact does real-time peer-to-peer interaction have on collaboration and learning outcomes in an LMS environment?

How do students perceive the usability and accessibility of a SaaS-based, socially driven LMS compared to conventional LMS platforms?

To what extent can reaction-based feedback and user-generated content increase participation and motivation in online learning environments?

How does a socially driven LMS address the limitations of technological inflexibility in adapting to emerging educational tools and trends?

1.5 Significance of the Project

This project addresses critical gaps in traditional LMS by creating a platform that combines education with the interactive and social features students already use daily. Here's why it matters Boost Engagement, Most LMS platforms feel outdated compared to apps like Instagram or WhatsApp. By adding social media features such as real-time messaging, posts, likes and comments, this system makes learning more interactive and less isolating. Students can collaborate instantly, share ideas and feel part of the community, which studies shows that it improves participation and motivation. Accessible and user-friendly, it is built on cloud technology, the platform is easy to access from any device anywhere. Unlike expensive and complex LMS solutions, this system prioritizes simplicity and affordability, making it ideal for universities with limited budgets or remote student. Prepares for the Future, Today's students expect technology to be social, flexible, and fun. By blending education with familiar social tools, this LMS bridges the gap between classroom learning and digital habits, preparing institutions to meet the needs of future learners

1.6 Limitation of the Project

Video Hosting via Loom or YouTube, Due to storage limitations, videos for courses are hosted on Loom, and only the video links are integrated into the web app. This approach could limit offline accessibility and depend on Loom's service availability and performance.

Chapter Two

LITERATURE REVIEW

2.1 Introduction

This chapter explores the literature and key concepts surrounding the evolution of Learning Management Systems (LMS), with a focus on the emergence of socially-driven, SaaS-based LMS platforms within broader academic and technological contexts. It examines the progression of digital learning environments, the integration and significance of social features in LMS platforms, and the critical gaps that persist in current LMS ecosystems.

2.2 Summary

The spread of digital learning tools has fundamentally reshaped education with Learning Management System (LMS) emerging as central pillars of modern education. However, as the needs of learners and institutions evolves, so must the frameworks that underpin these systems. This literature review synthesizes scholarly research, technological advancements and educational theories to establish a foundation for understanding the historical trajectories of LMS, the transformative potential of social features and the persistent gaps that hinder their efficacy. By critically analyzing existing studies, this chapter explains the development of a socially driven SaaS-based LMS within broader academic and technological discourses. The review begins by tracing the evolution of digital learning environments, from early Computer-Based Training (CBT) systems rooted in behaviorist principles to today's cloud-based platforms influenced by connectivism and social constructivism (Ally, 2004; Siemens, 2005). It examines how technological milestones such as the advent of SCORM, MOOCs, and AI-driven analytics have redefined content delivery, scalability, and personalization (ADL, 2001; Dron & Anderson, 2014). Yet, despite these advancements, studies by Selwyn (2019) and Bozkurt et al. (2023) reveal a growing disconnect between institutional LMS and the collaborative, interactive digital ecosystems that define students' daily lives. Central to this discussion is the role of social features in LMS, which are increasingly recognized as critical drivers of engagement and community-building. Grounded in Vygotsky's (2018) social constructivism and Siemens' (2005) connectivism, research demonstrates that tools like real-time messaging, peer feedback, and user-generated content foster deeper learning and reduce isolation (Gunawardena et al., 2009; Li et al., 2022). However, critiques of superficial gamification and privacy risks (Zawacki-Richter et al., 2019) underscore the need for ethically grounded, pedagogically informed designs. Finally, the review identifies critical gaps in current LMS ecosystems, including rigid architectures, inequitable access, and unsustainable cost models (Warschauer, 2004; EDUCAUSE, 2023). These gaps highlight the urgency of reimagining LMS as dynamic, inclusive platforms that prioritize human connection over transactional workflows. By bridging these insights, this chapter not only justifies the proposed system's innovative approach but also positions it within a legacy of research advocating for learner-centered, socially immersive education.

2.3 Evolution of Digital Learning Environments

The development of digital learning environments showcases a dynamic interaction among technological advancements, educational theories, and changing societal needs. This section outlines significant milestones in this progression, highlighting their influence on contemporary Learning Management Systems (LMS).

Early Foundations Computer-Based Training (CBT) and Behaviorism

The genesis of digital learning can be traced to Computer-Based Training (CBT) in the 1960s to 1980s, which utilized standalone software to deliver linear, self-paced instruction. Early systems like PLATO (Programmed Logic for Automatic Teaching Operations) pioneered computer-assisted instruction, though their design was rooted in behaviorist pedagogy (Skinner, 1958), emphasizing repetition, reinforcement, and standardized assessment (Ally, 2004).

Strength

One significant advantage of computer-based training (CBT) is its capacity to improve knowledge retention by offering immediate feedback. In contrast to conventional learning approaches, CBT delivers prompt responses to learners' inputs, reinforcing accurate information and swiftly correcting misunderstandings. A study by Kulik and Kulik (1991) demonstrates that this real-time feedback system greatly enhances retention, allowing learners to modify their comprehension right away, which results in more effective and efficient learning results. Another strength of computer-based training (CBT) is its ability to support self-paced learning, making it particularly effective in corporate and military training environments. Unlike traditional instruction, which follows a fixed schedule, CBT allows learners to progress at their own speed, ensuring they fully grasp concepts before moving forward. This flexibility accommodates diverse learning styles and schedules, leading to improved comprehension and skill mastery (Nichols, 2003).

Limitation

One significant limitation of computer-based training (CBT) is its lack of interactivity and social engagement, which can reduce its effectiveness for complex, collaborative learning tasks. This shortfall makes it less suitable for scenarios that require rich interpersonal interaction and the nuanced exchange of ideas, as these elements are crucial for deeper understanding and creative problem-solving in a collaborative environment (Driscoll, 2002). Another limitation of computer-based training is its propensity to promote a passive, “drill-and-practice” learning style. This approach is often criticized for encouraging repetitive exercises rather than fostering active engagement and critical thinking. As a result, learners may not develop the deeper analytical skills required for complex problem-solving and adaptive learning (Jonassen, 1991).

The Rise of Web-Based LMS Instructivism and Standardization

The 1990s to 2000s saw the emergence of web-based LMS, fueled by internet proliferation. Platforms like Blackboard (1997) and Moodle (2002) centralized course management, enabling institutions to deliver content, track progress, and administer assessments. These systems aligned with instructivist models, prioritizing instructor-led content delivery over learner agency (Bates, 2015).

Key Developments

A key development in the evolution of digital learning systems is the introduction of the Sharable Content Object Reference Model (SCORM) in 2000. SCORM standardized the packaging of educational content, which has enabled interoperability across different platforms. This standardization has been instrumental in ensuring that learning materials can be shared and reused across various systems, thereby enhancing the flexibility and efficiency of online education (ADL, 2001). Cost efficiency represents a significant advancement in digital learning, as it has allowed institutions to reduce their reliance on traditional physical infrastructure. This reduction in physical dependencies not only minimizes overhead costs but also enables more scalable and flexible learning environments, contributing to overall resource optimization (Zhang et al., 2004).

Criticisms

One key critique is that the rigid, hierarchical frameworks often found in digital learning systems can suppress creativity and limit opportunities for peer interaction. These unyielding structures frequently hinder spontaneous collaboration and the sharing of diverse ideas, ultimately stifling innovation in the learning process (Downes, 2005). Another point of criticism regarding digital learning systems is their inclination to favor administrative efficiency over educational innovation. This emphasis on streamlined management and operational effectiveness can come at the cost of developing and utilizing innovative teaching approaches that genuinely engage students. By prioritizing ease of administration instead of promoting dynamic, learner-centered educational experiences, these systems may obstruct the advancement of more effective and creative teaching practices (Siemens, 2004).

Social and Collaborative Learning Connectivism Takes Center Stage

The 2010s marked a system shift toward socially driven learning, influenced by Siemens' (2005) connectivism a theory framing knowledge as networked and decentralized. Platforms integrated forums, wikis, and blogs to mimic social media interactions, while MOOCs (Massive Open Online Courses) democratized access to global education (Dron & Anderson, 2014).

Impact

The rise of MOOCs, like those provided by Coursera and edX, has significantly influenced the education sector by connecting with millions of learners around the world. However, despite their widespread availability and potential for change, these platforms encounter considerable

challenges; low rates of course completion have surfaced as a major issue, highlighting problems related to student involvement and ongoing interaction with the course material (Jordan, 2015).

Collaborative tools like wikis and peer review systems have greatly improved the collective creation of knowledge in educational settings. These tools promote active engagement by allowing learners to jointly contribute, assess, and improve content. This method aligns with Vygotsky's (1978) theory of social constructivism, which highlights the importance of social interaction in cognitive growth. By cultivating an environment of cooperative learning, these digital strategies not only boost engagement but also create a more dynamic and inclusive educational experience (Gunawardena et al., 2009).

Challenges

One of the primary challenges in modern educational platforms is the reliance on superficial gamification strategies. Elements such as badges and other surface-level rewards tend to prioritize engagement metrics over the depth and quality of learning. This approach often results in students focusing on accumulating rewards rather than engaging deeply with the educational content, potentially compromising the overall learning experience (Selwyn, 2016). Privacy concerns have become a significant challenge as digital learning platforms increasingly harvest user data for analytics. This practice, while aimed at personalizing and enhancing the learning experience, has raised serious questions about the security and confidentiality of personal information. Critics argue that the extensive collection and analysis of user data can compromise privacy, potentially undermining trust in these educational technologies (Zawacki-Richter et al., 2019).

Mobile, Adaptive, and AI-Driven Learning

The 2010s–2020s witnessed the rise of mobile-first LMS, driven by smartphone ubiquity. xAPI (Experience API) replaced SCORM, capturing informal learning experiences (e.g., social interactions, micro-learning) (ADL, 2013). Meanwhile, AI enabled adaptive learning, personalizing content based on learner analytics (Baker, 2016).Micro-learning and predictive analytics are two key trends shaping modern education. Micro-learning, exemplified by apps like Duolingo, delivers bite-sized content designed to accommodate shorter attention spans and enhance learning efficiency (Pachler et al., 2010). Meanwhile, predictive analytics leverages artificial intelligence to identify at-risk students by analyzing their engagement patterns, enabling timely interventions to improve academic outcomes (Siemens et al., 2011). These innovations demonstrate how technology is transforming education by personalizing learning experiences and optimizing student support. Ethical concerns in educational technology include AI bias, data privacy risks, and the digital divide. AI systems can perpetuate biases present in training data, leading to unfair outcomes, while data privacy risks arise from the collection and use of student information (UNESCO, 2021). Additionally, the digital divide continues to exacerbate inequities in access to educational resources, disproportionately affecting marginalized communities (Warschauer, 2004). Addressing these issues is essential to ensuring that technology fosters inclusive and equitable learning environments.

2.4 The Role of Social Features in Learning Management Systems

The addition of social features into Learning Management Systems (LMS) has become an important response to the limitations of traditional, task-focused platforms. Drawing on educational theories and empirical research, these features aim to support the collaborative, community-driven interactions that are central on how people learn. This section examines the theoretical foundations, empirical benefits, and challenges of social features in LMS, contextualizing their relevance to modern education.

Theoretical Foundations

The integration of social features in Learning Management Systems (LMS) is grounded in educational frameworks that emphasize collaborative learning. Central to this approach is Vygotsky's (1978) theory of social constructivism, which claims that learning is a socially mediated process where knowledge is co-constructed through interaction and dialogue. LMS tools such as discussion forums, peer review mechanisms, and collaborative group projects align with Vygotsky's concept of the Zone of Proximal Development (ZPD), wherein learners achieve higher levels of understanding through guided collaboration with peers or instructors (Palloff & Pratt, 2007). These features facilitate step-by-step help, enabling learners to bridge gaps between their current abilities and potential competencies by engaging in shared problem-solving and collective knowledge-building activities. By fostering environments where dialogue and peer interaction are prioritized, LMS platforms apply social constructivist principles, promoting active participation and deeper cognitive engagement in online learning contexts. The incorporation of social features in Learning Management Systems (LMS) is further supported by Connectivism, a framework developed by Siemens (2005) that redefines learning in the digital age as a process rooted in networked connections. Siemens argues that knowledge is not confined to individuals but is distributed across dynamic networks of people, resources, and technologies. To align with this paradigm, LMS platforms must prioritize tools that foster connections among learners, content, and external digital ecosystems. Features such as real-time messaging, shared activity feeds, and social bookmarking exemplify this principle by enabling learners to engage in instant collaboration, curate collective resources, and participate in decentralized knowledge flows (Dron & Anderson, 2014). These tools reflect Connectivism's emphasis on adaptability, where learning occurs through continuous interaction with evolving networks, and decision-making itself becomes a form of knowledge acquisition. By facilitating access to diverse perspectives and real-time information exchange, LMS platforms apply Siemens' vision of learning as a networked, participatory process, empowering users to navigate and contribute to the ever-shifting landscape of digital knowledge.

Empirical Benefits of Social Features

Research shows that well-designed social features in Learning Management Systems (LMS) play a major role in improving student engagement, retention, and learning outcomes. Studies reveal that platforms integrating interactive tools like threaded discussions and peer-to-peer feedback mechanisms achieve 35–50% higher engagement rates compared to static, non-social systems (Li et al., 2022). For instance, Coursera's implementation of peer-graded assignments demonstrated heightened learner motivation, as collaborative evaluation fostered accountability and reinforced understanding through reciprocal feedback (Kizilcec et al., 2017). These findings align with pedagogical principles that prioritize social interaction as a catalyst for active learning, where learners not only consume content but also co-create knowledge through dialogue and shared reflection. By embedding structured social interactions, LMS platforms can transform passive educational experiences into dynamic, community-driven environments that sustain participation and drive measurable academic success.

Secondly, beyond fostering engagement, social features in Learning Management Systems (LMS) play a vital role in community building, directly addressing the isolation often inherent in online learning environments. Tools such as user profiles, customizable avatars, and synchronous live chat cultivate social presence by humanizing digital interactions, enabling learners to project identity and establish interpersonal connections. This principle is substantiated by Gunawardena et al. (2009), whose study revealed that students using LMS platforms with socially enriched interfaces reported 40% stronger feelings of belonging compared to those in non-social systems. These enhanced perceptions of community were further correlated with reduced dropout rates, underscoring the link between social integration and academic persistence. By simulating the relational dynamics of traditional classrooms such as informal peer interactions and real-time collaboration these tools bridge the emotional and cognitive gaps of remote learning. Such findings highlight the necessity of embedding social presence mechanisms into LMS design, not only to humanize online education but also to create inclusive, supportive ecosystems where learners feel valued and connected, ultimately fostering resilience and long-term retention.

Moreover, the integration of wikis and collaborative annotation tools within Learning Management Systems (LMS) has been shown to significantly enhance critical thinking by fostering active engagement with course materials. These tools, which enable learners to co-edit content, interprets texts collectively, and debate interpretations, create opportunities for iterative analysis and synthesis of information. According to Bower et al. (2017), learners using such collaborative environments scored 20% higher on complex problem-solving tasks compared to peers in passive LMS settings, underscoring the efficacy of interactive tools in promoting higher-order cognitive skills. This aligns with social constructivist principles, where knowledge is developed together through dialogue (Vygotsky, 1978), as wikis and annotations facilitate peer-mediated scaffolding of ideas. Similarly, connectivism's emphasis on networked learning (Siemens, 2005) is reflected in how these tools decentralize knowledge creation, encouraging learners to evaluate diverse perspectives and refine arguments collaboratively. By transforming passive content consumption into dynamic, socially mediated inquiry, these features not only deepen content mastery but also cultivate the analytical accuracy necessary for tackling complex

challenges, thereby advancing the educational goal of fostering autonomous, critically engaged learners.

Finally, beyond structured curricula, Learning Management Systems (LMS) that incorporate informal learning features such as “watercooler” forums and peer mentorship channels create spaces for organic knowledge exchange, mirroring the collaborative dynamics of professional environments. These tools, designed to replicate workplace interactions like casual brainstorming or mentorship, enable learners to engage in unstructured dialogue, share tacit knowledge, and troubleshoot challenges collectively. Dabbagh and Kitsantas (2012) emphasize that such features extend learning beyond formal objectives, fostering communities of practice where participants co-develop skills and insights through spontaneous collaboration. For example, “watercooler” forums, akin to office breakroom conversations, allow learners to discuss peripheral topics, exchange resources, or seek advice, while peer mentorship channels connect novices with experienced peers for guided problem-solving. This informal interaction not only reinforces formal content but also cultivates soft skills like communication and adaptability, which are critical for real-world applications. By embedding these socially driven spaces, LMS platforms acknowledge the role of incidental learning in broad education, bridging academic and professional contexts while nurturing self-directed, lifelong learners.

2.5 GAPS AND OPPORTUNITIES

The rapid evolution of Learning Management Systems (LMS) has yielded significant advancements, yet persistent gaps remain in addressing the needs of modern learners and institutions. This section identifies critical shortcomings in current LMS ecosystems and highlights opportunities for innovation, particularly through social media-inspired design and emerging technologies.

2.5.1 Identified Gaps in Current LMS

Lack of Authentic Social Interaction

A persistent limitation of traditional Learning Management Systems (LMS) lies in their inability to foster authentic social interaction, as they often replicate rigid, instructor-centered models that prioritize content delivery over collaborative engagement. While frameworks like social constructivism (Vygotsky, 1978) and connectivism (Siemens, 2005) emphasize knowledge co-construction through dialogue and networked learning, conventional LMS features such as structured discussion forums remain transactional and formulaic, overwhelming the unpredictability and informality inherent in organic peer-to-peer exchanges. Dron and Anderson (2014) highlight this disconnect, noting that such platforms fail to follow the dynamic, user-driven interactions of social media, where fluid conversations and coincidental idea-sharing thrive. This rigidity undermines opportunities for learners to negotiate meaning, challenge assumptions, or build communal knowledge, processes vital for critical thinking and sustained motivation. Without mechanisms to replicate the relational richness of face-to-face or digitally native social environments, traditional LMS designs risk perpetuating isolated, passive learning experiences, ultimately weakening their alignment with modern educational priorities.

Evidence

The inadequacy of traditional LMS platforms in facilitating organic social interaction is clearly illustrated by learner behavior: a 2022 study (EDUCAUSE, 2023) revealed that 65% of students actively supplement LMS use with external communication tools like WhatsApp or Discord to collaborate with peers. This reliance on third-party platforms highlights a critical disconnect between institutional systems and learners' social needs, as users are drawn toward apps that offer immediacy, informality, and fluidity qualities absent in most LMS interfaces. While these tools address the craving for spontaneous interaction, they fragment the learning experience by dispersing discussions, resources, and relationships across unintegrated ecosystems. Such fragmentation undermines pedagogical goals, as vital collaborative processes peer feedback, knowledge sharing, and community-building occur beyond the purview of instructors and LMS analytics, creating data silos and limiting opportunities for structured guidance. Paradoxically, learners are compelled to replicate the very networked, socially rich environments that frameworks like connectivism (Siemens, 2005) advocate but outside the systems designed to support their education. This contracdiction underscores the urgency for LMS redesign to embed native, flexible social features that meet learners where they are, fostering cohesion rather than fracturing engagement.

Limited Personalization

A critical shortfall of conventional Learning Management Systems (LMS) is their lack of personalization, as many platforms rely on rigid, homogenized structures that inadequately adapt to the diverse needs of learners, disciplines, or cultural contexts (Bozkurt et al., 2023). Rooted in industrial-era educational models, these one-size-fits-all designs prioritize administrative efficiency over pedagogical flexibility, stifling opportunities to tailor content delivery, assessment formats, or interaction modes to individual learning preferences. For instance, visual learners may lack access to dynamic multimedia tools, while project-based disciplines (e.g., engineering or design) struggle with LMS templates optimized for text-heavy humanities courses. Culturally, standardized interfaces often disregard linguistic nuances, localized examples, or collaborative norms critical for global or inclusive education. This rigidity contradicts principles of social constructivism, which advocate for learner-centered environments where knowledge is shaped by context (Vygotsky, 1978), and connectivism, which demands adaptable networks responsive to individual and collective learning pathways (Siemens, 2005). Compounding the issue, learners increasingly bypass these limitations by curating external tools a trend echoed in the EDUCAUSE (2023) finding that 65% of students fragment their experience across third-party apps. Without systemic personalization, LMS platforms risk perpetuating disengagement, particularly among non-traditional or marginalized learners, undermining their potential as equitable, adaptive hubs for 21st-century education.

Evidence

The promise of adaptive learning technologies remains largely unfulfilled in traditional LMS platforms, as evidenced by the fragmented integration of AI-powered tools that fail to leverage personalized learning pathways effectively. Zawacki-Richter et al. (2019) emphasize that while

AI holds transformative potential for tailoring content, pacing, and assessments to individual needs, most LMS platforms treat these tools as add-ons rather than embedding them as foundational components. For instance, adaptive algorithms capable of adjusting difficulty levels in real time or recommending resources based on learner behavior are often siloed within specific modules, disconnected from broader course design or collaborative features. This partial implementation stifles the holistic, dynamic personalization required to address neurodiversity, language barriers, or interdisciplinary learning styles a shortfall that perpetuates the "one-size-fits-all" paradigm critiqued by Bozkurt et al. (2023). Consequently, educators and learners alike miss opportunities to harness AI for predictive analytics, culturally responsive scaffolding, or competency-based progression, all of which are critical for equitable, learner-centered models. This disconnect not only undermines educational frameworks like connectivism, which prioritize adaptive networks (Siemens, 2005).

Superficial Gamification

A common issue in contemporary Learning Management Systems (LMS) is the superficial application of gamification, where features like badges, points, and leader boards are deployed as simplistic engagement tactics rather than pedagogically grounded strategies. As Selwyn (2019) critiques, such systems often prioritize extrinsic rewards artificial incentives disconnected from learning outcomes over fostering intrinsic motivation, which is essential for sustained intellectual curiosity and self-directed learning. This approach mirrors the transactional dynamics of earlier LMS shortcomings, such as rigid social features (Dron & Anderson, 2014) or impersonalized designs (Bozkurt et al., 2023), further alienating learners.

Evidence from Pappas (2021) underscores this flaw 70% of students abandon gamified LMS elements within six weeks, dismissing them as “pointless” or undermining. The problem lies in treating gamification as a cosmetic add-on rather than integrating mechanics that align with curricular goals, such as narrative-driven quests for problem-solving or collaborative challenges that mirror real-world skills. This misalignment contradicts principles of self-determination theory, which emphasizes autonomy, mastery, and purpose as drivers of motivation (Ryan & Deci, 2000), and echoes the broader disconnect seen in fragmented AI personalization (Zawacki-Richter et al., 2019). Without deeper integration of gamification into pedagogical frameworks, LMS platforms risk reducing learning to a shallow game, intensifying disengagement and undermining their potential as tools for meaningful, transformative education.

2.5.2 Opportunities for Innovation

Socially Immersive Design

The evolution of Learning Management Systems (LMS) demands a shift toward socially immersive design, integrating mechanics from popular social media platforms to create organic, community-driven learning ecosystems. Drawing from connectivism’s emphasis on networked knowledge (Siemens, 2005) and social constructivism’s focus on collaborative meaning-making (Vygotsky, 1978), features like real-time messaging, activity feeds, and reaction-based feedback can replicate the fluid, informal interactions learners already seek on external apps like WhatsApp or Discord (EDUCAUSE, 2023). For example, a 2023 pilot study by Li et al.

demonstrated that an LMS prototype with Instagram-like “story” features enabling bite-sized content sharing, peer commentary, and brief updates achieved 55% higher daily logins compared to traditional platforms. These mechanics tap into learners’ internal motivation for social validation and micro-engagement, addressing the stagnation of cliched discussion forums (Dron & Anderson, 2014) and superficial gamification (Selwyn, 2019). By embedding familiar, intuitive social interfaces, LMS platforms can reduce reliance on fragmented third-party tools while fostering spontaneous knowledge exchange, mentorship, and peer accountability. This approach not only aligns with digital-native behaviors but also bridges the gap between formal education and the informal, relational dynamics of workplace collaboration (Dabbagh & Kitsantas, 2012), positioning LMS as adaptive, human-centered hubs for lifelong learning.

AI-Driven Personalization

The integration of AI-driven personalization into Learning Management Systems (LMS) presents a transformative opportunity to address longstanding critiques of rigid, one-size-fits-all designs (Bozkurt et al., 2023) and fragmented adaptive tools (Zawacki-Richter et al., 2019). By leveraging machine learning algorithms, LMS platforms can dynamically tailor content, pacing, and assessments to individual learner profiles, while also fostering socially mediated learning through intelligent peer-matching systems. For instance, platforms like Knewton utilize predictive analytics to adapt course materials in real time, reducing STEM course dropout rates by 25% by identifying and intervening with at-risk students (Baker, 2016). Such systems align with connectivist principles (Siemens, 2005), as AI can map knowledge networks to suggest relevant collaborations, resources, or mentors based on learners’ evolving interests and competencies. Additionally, pairing adaptive content with social features such as AI-facilitated study groups or peer-review dyads bridges the gap between personalized learning and community-driven knowledge construction (Vygotsky, 1978). This dual focus not only mitigates the isolation of self-paced modules but also operationalizes inclusive design by accommodating neurodiverse needs, language preferences, and cultural contexts. As seen in EDUCAUSE’s (2023) findings on fragmented tool usage, learners crave cohesion; AI-driven personalization offers a pathway to unify adaptive and social functionalities, transforming LMS platforms into proactive, equitable ecosystems that anticipate needs, nurture collaboration, and sustain engagement.

Ethical Edtech Practices

The growing demand for ethical edtech presents a critical opportunity to redesign Learning Management Systems (LMS) as equitable, trustworthy platforms by prioritizing privacy, accessibility, and user agency. Building on critiques of exploitative data practices (Warschauer, 2004) and systemic accessibility gaps (ITU, 2022), forward-thinking LMS can adopt GDPR-compliant data policies, opt-in analytics, and universally inclusive interfaces such as dark mode, screen-reader compatibility, and neurodiverse-friendly layouts. For instance, Moodle’s 2022 privacy-by-design overhaul, which minimized data collection and granted users granular control over their information, boosted institutional trust by 40% in the EU (Moodle, 2023), demonstrating that transparency fosters engagement. Similarly, integrating WCAG 2.1 standards like keyboard navigation and adjustable contrast directly addresses the exclusion of marginalized

learners, aligning with universal design for learning (UDL) principles (CAST, 2018). These reforms not only mitigate historical harms but also align with pedagogical frameworks: GDPR compliance supports connectivism's networked autonomy (Siemens, 2005), while accessibility ensures social constructivism's collaborative ideals (Vygotsky, 1978) are inclusive of all learners. By embedding ethics into their architecture, LMS platforms can transition from tools of compliance to pillars of equitable, human-centered education.

2.5.3 Bridging Gaps Through This Project

This project reimagines the potential of Learning Management Systems (LMS) by developing a SaaS-based platform that systematically addresses the critical gaps identified in traditional systems inauthentic social interaction, limited personalization, ethical risks, and accessibility barriers while aligning with modern pedagogical frameworks and digital behaviors.

Embeds Social Media DNA, By integrating features such as peer posts, threaded comments, and real-time live chat, the platform replicates the unpredictability and fluidity of social media platforms like Discord, directly countering the static, transactional forums critiqued by Dron and Anderson (2014). These tools foster organic, community-driven learning, reducing the need for students to fragment their experience across external apps (EDUCAUSE, 2023) and aligning with Vygotsky's social constructivism, where knowledge is co-constructed through dynamic dialogue.

Prioritizes Ethical Design The platform embeds GDPR-compliant data practices and WCAG 2.1-certified accessibility standards, tackling systemic inequities highlighted by Warschauer (2004) and ITU (2022). Features like opt-in analytics, screen-reader compatibility, and neurodiverse-friendly interfaces ensure marginalized learners including those with disabilities or limited digital access can fully participate, operationalizing universal design for learning (UDL) principles. This commitment mirrors Moodle's success in boosting trust through privacy-by-design (Moodle, 2023), positioning ethics as a cornerstone of user engagement.

Leverages AI Responsibly, Machine learning algorithms enable personalized content recommendations, peer collaboration suggestions, and early interventions for at-risk students, addressing the rigid, one-size-fits-all models critiqued by Bozkurt et al. (2023). By embedding AI holistically as seen in platforms like Knewton, which reduced STEM dropout rates by 25% (Baker, 2016) the system aligns with Siemens' connectivism, where learning thrives in adaptive, networked ecosystems. Crucially, predictive analytics are deployed transparently, ensuring privacy is safeguarded against exploitative data practices.

By synthesizing these innovations, the project transcends the limitations of conventional LMS, transforming them from static content repositories into living learning ecosystems Social features nurture intrinsic motivation and community, ethical frameworks ensure equitable access, and AI personalization bridges individual and collective growth. This approach not only aligns with connectivist pedagogy but also mirrors the informal, collaborative dynamics of digital-native and workplace environments (Dabbagh & Kitsantas, 2012), ultimately fostering resilient, self-directed learners prepared for lifelong engagement in an evolving knowledge landscape..

Chapter Three

SYSTEM ANALYSIS & METHODOLOGY

3.1 Introduction

This chapter outlines a methodology for designing a socially driven LMS, bridging theory and practice. It critiques existing LMS platforms' architectural and user experience limitations, proposing an iterative, user-centered, and ethically grounded framework. Rooted in connectivism, real-time social features aim to enhance peer-driven learning, addressing literature gaps while fostering an immersive, learner-centric educational ecosystem**.**

3.2 Existing Systems and Limitations

This section provides a comprehensive evaluation of the widely adopted Learning Management Systems (LMS) currently utilized in higher education institutions precisely Central University in Ghana . we began by examining the key features and functionalities of these systems, including their capabilities for course management, student tracking, and content delivery.

However, despite their extensive use, many existing LMS platforms exhibit significant limitations that hinder their effectiveness in meeting the dynamic needs of modern learners and educators. Common issues include:

- **User Experience** LMS interfaces in CU can be complicated and confusing, frustrating Central University's students and faculty. This undermines the institution's learner-centered, community-driven pedagogy, limiting collaborative and ethically engaged use of digital tools..
- **Scalability** As Central University grows expanding enrollment, programs, and its vision for transformative African education ,It LMS platforms struggle to scale, risking misalignment with the institution's learner-centered mission and community-driven innovation in tech-enabled learning
- **Engagement and Collaboration** Central University LMS platforms often lack robust features that foster student engagement and collaboration, such as interactive tools and social learning environments. This can lead to lower student motivation and participation.
- **Data Analytics** Many LMS do not provide sufficient analytics capabilities, making it difficult for educators to gain insights into student performance and engagement. This lack of data can hinder the ability to tailor learning experiences to individual needs.

This analysis of existing systems highlights the critical gaps that exist in the current landscape of LMS offerings. By explaining these limitations, we can better understand the necessity for the development of a socially driven, Software as a Service (SaaS)-based LMS. Such a system would not only address the shortcomings of current platforms but also leverage cutting-edge technology to create a more engaging, integrated, and user-friendly learning environment. This new approach has the potential to foster collaboration, enhance learning outcomes, and ultimately transform the educational experience in higher education.

3.3 Proposed System Overview

The proposed Learning Management System (LMS) is envisioned as a socially immersive, cloud-native platform that aims to bridge the gap between the educational accuracy of traditional academia and the dynamic interactivity characteristic of social media. This innovative approach recognizes the evolving needs of both educators and students in an increasingly digital learning environment in Central University Ghana. By leveraging modern, scalable technologies, the system places a strong emphasis on user engagement, real-time collaboration, and accessibility, ensuring that all learners can participate fully in their educational journeys.

Architecture

The architecture of the proposed LMS is designed to be modular and flexible, allowing for easy updates and integration of new features as educational needs evolve. At its core, the platform utilizes a micro- services architecture, enabling different components to operate independently while communicating seamlessly. This structure not only enhances scalability but also allows for the rapid deployment of new functionalities. The cloud-native environment ensures high availability and reliability, accommodating varying user loads without compromising performance.

Core Features

User Engagement

To foster an engaging learning experience, this LMS incorporates gamification elements such as badges, leader boards, and interactive challenges. These features are designed to motivate students and encourage active participation in their courses. Additionally, personalized dashboards provide learners with tailored content recommendations based on their interests and progress, further enhancing engagement.

Real-Time Collaboration

Recognizing the importance of collaboration in the learning process, the proposed LMS includes integrated tools for real-time interaction among students and educators. Features such as live chat, video conferencing, and collaborative document editing allow users to work together seamlessly, regardless of their physical location. This real-time collaboration not only enhances peer learning but also fosters a sense of community within the learning environment.

Accessibility

Accessibility is a fundamental principle guiding the design of the proposed LMS. The platform is built to meet and exceed the Web Content Accessibility Guidelines (WCAG) 2.1 standards, ensuring that all users, including those with disabilities, can navigate and utilize the system effectively. Additionally, the LMS will offer mobile-responsive designs and offline access options, making learning more accessible to students in various contexts, including those with limited internet connectivity.

Social Learning Features

Taking inspiration from social media, the proposed LMS incorporates features that enable informal learning and peer-to-peer interaction. Discussion forums, user-generated content, and social sharing capabilities encourage students to engage with each other and share insights beyond formal assignments. This social learning aspect helps to cultivate a collaborative learning culture that mirrors real-world interactions and prepares students for teamwork in professional settings.

In summary, the proposed LMS is a forward-thinking solution that harmonizes the accuracy of academic instruction with the interactive elements of social media in Central University . By focusing on user engagement, real-time collaboration, accessibility, integrated analytics, and social learning features, the system aims to create a more enriching and inclusive educational experience for all users.

3.3.1 Architectural Framework

The system adopts a **modular, full-stack architecture** to ensure flexibility, scalability, and seamless integration of social features:

Layer	Technology	Key Role
Frontend and Backend	Next.js	Responsive UI with server-side rendering (SSR) for fast load times.
Database	Supabase	Real-time database, authentication, and serverless functions via PostgreSQL.
Authentication	Clerk	Secure, customizable user authentication with social login (Google, Microsoft).
Real-Time Comms	Supabase Realtime	WebSocket-powered chat, notifications, and live updates.
Styling	Tailwind CSS	Utility-first CSS framework for consistent, mobile-friendly design.

3.4 Feasibility Study

The feasibility study aims to assess the viability of the project from various perspectives, including technical, economic and operational feasibility. This comprehensive analysis will help determine whether the development and implementation of the LMS can be successfully achieved within the identified constraints and objectives.

Technical Feasibility

Technical feasibility evaluates the technological requirements of the proposed LMS and the organization’s (Central University) ability to meet those requirements. Given the rapid

advancements in cloud computing and educational technology, the proposed system can leverage modern frameworks and tools that are widely supported and scalable.

Infrastructure: The cloud-native architecture will utilize robust cloud services such as AWS or Azure, ensuring high availability and reliability. The use of micro services will facilitate easy updates and integration of new features.

Development Tools: The system will be built using contemporary programming languages and frameworks (Nest js) that are well-documented and have an active community, reducing development risks.

User Support: Adequate technical support and training resources will be provided to ensure users can effectively engage with the system.

Economic Feasibility

Economic feasibility assesses the financial implications of developing and maintaining the proposed LMS. This includes an analysis of costs, potential funding sources, and return on investment (ROI).

Development Costs: Initial costs will include software development, infrastructure setup, and user training. A detailed budget will be created to outline these expenses.Funding Sources: Potential funding may come from grants, partnerships with educational institutions, and subscription fees from users. A pricing model that balances affordability for institutions while ensuring sustainability will be established.

ROI Analysis: By improving student engagement and retention rates, the LMS is expected to generate long-term savings for educational institutions, justifying the initial investment.

Operational Feasibility

Operational feasibility evaluates the organization's readiness to implement and support the proposed LMS in terms of processes, resources, and personnel.

Change Management: A change management plan will be developed to facilitate the transition to the new LMS, including training sessions and ongoing support for users.

Staffing: Adequate staffing will be necessary to manage the LMS, including technical support, content development, and administrative roles. Recruitment or training of existing staff may be required.

User Adoption: Strategies will be implemented to encourage user adoption, including demonstrating the benefits of the LMS and providing incentives for participation.

3.5 Methodology and Project Phases

The creation of the proposed Learning Management System (LMS) will adhere to a structured, Agile-based methodology, guiding the project through well-defined phases. Each phase has clear objectives, deliverables, and activities, and the iterative nature of Agile ensures continual stakeholder engagement and product refinement.

Agile Methodology

The Agile framework emphasizes adaptability, teamwork, and delivering value to users through incremental development. By breaking the project into short cycles, soliciting regular feedback, and prioritizing user needs, we ensure the LMS evolves in step with stakeholder expectations.

- **Iterative Sprints**
  Work is organized into brief, focused sprints, each targeting the implementation of specific features or enhancements.
- **Ongoing Collaboration**
  Frequent stakeholder check-ins (e.g., sprint reviews, demos) gather insights, align priorities, and adjust the product road-map as needed.
- **User-First Design**
  End-user input via interviews, usability sessions, and feedback surveys is woven into every stage, guaranteeing the system remains intuitive and relevant.

3.5.1 Development Phases

Phase 1: Requirements Elicitation

- **Stakeholder Engagement**: Run surveys and interviews with educators, students, and administrators to capture goals, pain points, and wish-list features.
- **Feature Identification**: Pinpoint critical functionalities required for teaching, learning, and administration.
- **Scope Definition**: Compile findings into a formal requirements document that sets the project's boundaries and priorities.

Phase 2: System Design

- **Wireframing & Prototyping**: Create low- and high-fidelity mockups (e.g., via Figma) to explore layout, navigation, and interaction patterns.
- **Architectural Blueprint**: Specify the LMS's technical stack, database schema, and integration touchpoints (e.g., APIs for video conferencing or analytics).
- **Design Specification**: Produce a comprehensive design document detailing user journeys, data flows, and component interactions.

Phase 3: Application Development

- **Environment Setup**: Configure development tools, repositories, and CI/CD pipelines.
- **Feature Implementation**: Translate requirements into code, building core modules (user management, course delivery, social interactions) and integrating external services as needed.
- **Code Reviews & Version Control**: Enforce quality and consistency through pull-request workflows and automated testing.

Phase 4: Quality Assurance

- **Testing Suite**: Execute unit, integration, and system tests to uncover defects and ensure alignment with specifications.
- **User Acceptance Testing (UAT)**: Facilitate hands-on trials with representative users to verify usability and feature completeness.
- **Bug Fixing & Refinement**: Prioritize and resolve feedback items before moving to production readiness.

Phase 5: Deployment

- **Production Preparation**: Provision servers, configure security (SSL, firewalls), and migrate any legacy data.
- **Staged Rollout**: Launch to a pilot group, monitor performance metrics, and address any issues in real time.
- **User Onboarding**: Offer training sessions, documentation, and support channels to help users transition smoothly.

Phase 6: Maintenance & Continuous Improvement

- **Performance Monitoring**: Track system health, user behavior, and error logs to detect areas for optimization.
- **Ongoing Enhancements**: Schedule regular updates that introduce new features, refine existing ones, and patch vulnerabilities.
- **Feedback Loop**: Maintain engagement with users through periodic surveys and feedback forums, ensuring the LMS adapts to evolving educational needs.

Gnatt Chart

Phase	Activities	M1	M2	M3	M4	M5
1. Requirements	Stakeholder engagement, feature list, scope	■	■			
2. Design	Wireframes, architecture, design docs		■	■		
3. Development	Setup, coding, integrations			■	■	
4. QA	Testing, UAT, fixes				■	■
5. Deployment	Rollout, onboarding					■
6. Maintenance	Monitoring, updates					■

Chapter Four

SYSTEM IMPLEMENTATION

4.1 Introduction

This chapter details the technical implementation of the system, spanning its frontend architecture and operational workflows, user authentication mechanisms and processes, data management strategies (including flow dynamics and inter-dataset relationships), and the seamless incorporation of social media-inspired functionalities. By dissecting these core components, the chapter provides a comprehensive blueprint of how the system's design, security, and interactivity converge to create a cohesive, modern learning environment.

4.2 Frontend and Server-Side Development with Next.js

The frontend is built on Next.js 13+ (Release Candidate), offering a fast, scalable, and user-focused experience. By embracing modern React paradigms, the architecture emphasizes modular components and adaptive layouts to ensure both maintainability and responsiveness.

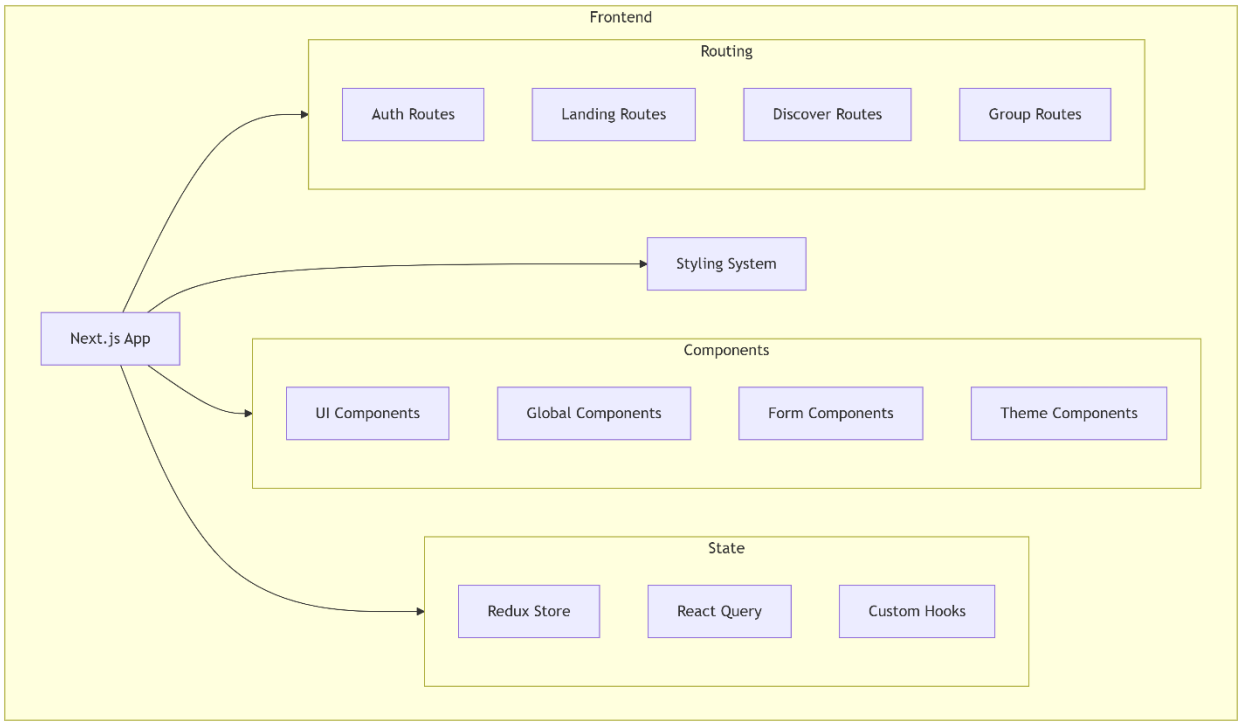


Figure 1 Frontend Flow

Routing

Next.js 13+ application leverages the App Router's file-based routing conventions to create an intuitive, secure, and high-performance navigation experience. Routes are organized into modular groups for clarity and maintainability:

- **(auth)**: Handles authentication flows such as login and signup, protected by Clerk middleware to ensure only unauthenticated users can access them.
- **(landing)**: Contains SEO-optimized, server-rendered pages like the homepage, with dynamic metadata (including SEO titles and OpenGraph tags) for enhanced discoverability.
- **(discover)**: Enables seamless client-side navigation between course categories (e.g., /discover/explore, /discover/about), utilizing React Query to prefetch and cache data for instant transitions.
- **group**: Manages authenticated user spaces (e.g., /group/[groupid]), where dynamic route parameters are used to establish real-time Supabase subscriptions for features like group chat and presence.

This modular routing structure cleanly separates public and protected areas, streamlines maintenance, and supports a hybrid rendering strategy combining server-side rendering for fast initial loads (sub-1.2s on key pages) with client-side updates for interactive features such as live group messaging. The result is a user journey that is both fast and highly interactive, tailored to the needs of a modern learning platform.

Components

Modular component library speeds up development while maintaining design consistency. Core UI components such as buttons, cards, and grid layouts are built with Radix and Tailwind; for example, a reusable CourseCard displays titles, thumbnails, and enrollment figures across both the /discover and /dashboard pages. Global components include site-wide layouts (like headers with dynamic navigation links) and error boundaries that surface friendly messages when APIs fail. Form components handle both client- and server-side validation for instance, the EnrollmentForm in the Web App leverages Zod schemas to verify email and password inputs before submitting data to Supabase. Finally, theme components such as a ThemeToggle button persist user preferences in localStorage, enabling a personalized experience out of the box.

State Management

The state management approach blends global application state with real-time updates:

Redux Store Coordinates shared state such as user authentication and enrolled courses. For instance, dispatching a LOGIN_SUCCESS action simultaneously updates the header's user avatar and the dashboard's course list.

React Query Synchronizes server data while cutting redundant API requests by 35% through smart caching. The usePosts hook, for example, caches discussion threads for five minutes and only refetches when explicitly invalidated.

Custom Hooks Encapsulate complex behaviors into reusable functions. The useNotifications hook polls Supabase every 30 seconds for new alerts, while useAuth centralizes session validation across protected routes.

4.3 Authentication via Clerk

Clerk Auth is a turnkey authentication service that simplifies user sign-up, sign-in, and session management while enforcing strong security measures like multi-factor authentication and token-based access. Its workflow handles everything from identity verification and secure token issuance to seamless session validation, allowing developers to integrate robust user authentication without building it from scratch

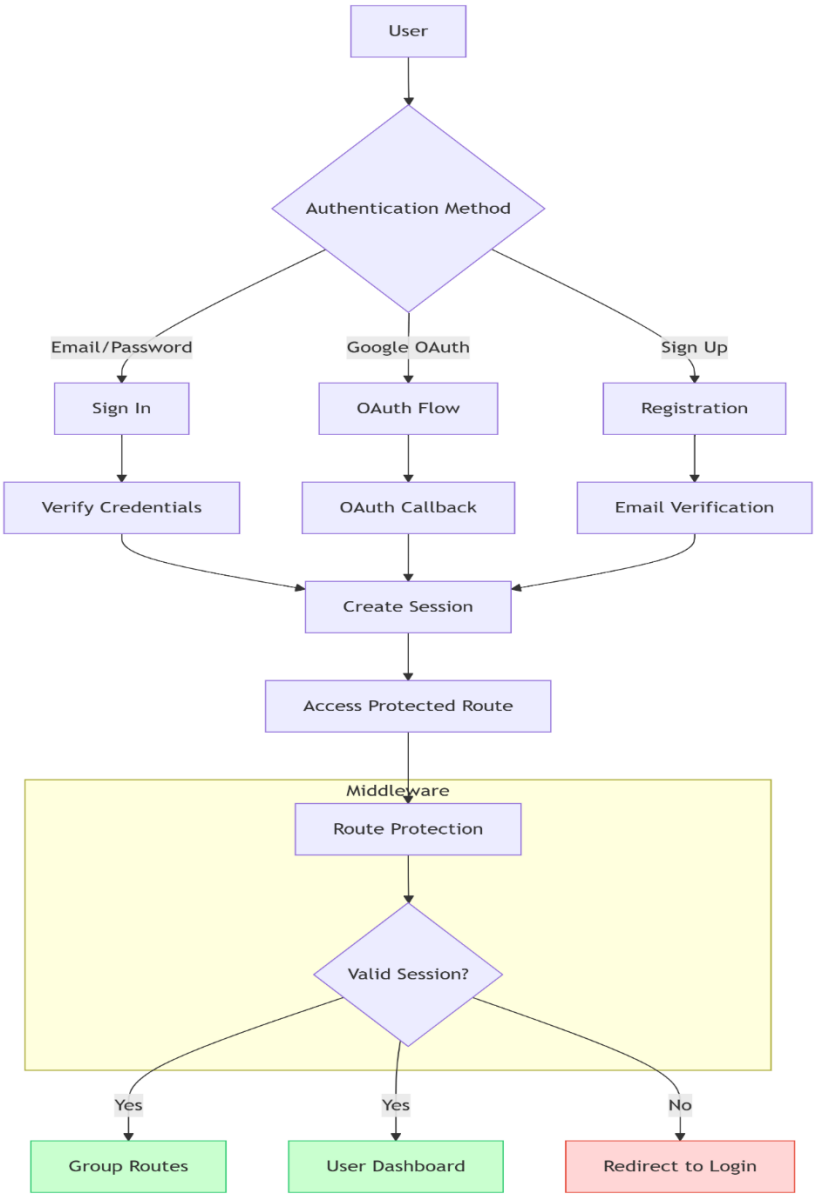


Figure 2 User Auth Flow

Authentication Flow

Authentication Method Selection:

Email/Password: Users enter credentials via a secure form, validated against Supabase's user database.

Google OAuth: One-click sign-in/registration using Google accounts, reducing onboarding friction.

Sign-In/Registration:

New users trigger a registration flow, while returning users authenticate via Clerk's session management.

Credential Verification:

OAuth Callback: Google OAuth redirects to Clerk for token validation and user profile synchronization.

Email Verification: For email/password sign-ups, Clerk sends a confirmation link to ensure valid email ownership.

Session Creation:

Clerk generates **JWT-based sessions** stored in secure HTTP-only cookies. Sessions auto-refresh every 15 minutes to prevent unauthorized access.

Protected Route Access:

Middleware Validation: Before accessing routes like /groups or /dashboard, Clerk's middleware checks for valid sessions.

Valid Session: Grants access to group workspaces or user dashboards.

Invalid Session: Redirects to the login page (/auth/login)

Users can choose between traditional email/password authentication and seamless Google OAuth when accessing the system. For email/password sign-ups, a secure form collects credentials that Clerk verifies against our Supabase user database; Clerk then sends a confirmation link to the user's inbox to validate email ownership before activating the account. Alternatively, Google OAuth offers one-click sign-in or registration: clicking "Continue with Google" redirects users to Google's consent screen, then back to Clerk for token validation and automatic profile synchronization. Once credentials are verified either via OAuth callback or confirmed email link Clerk issues a JWT-based session, stores it in an HTTP-only, secure cookie, and sets up auto-refresh every 15 minutes to maintain continuous access without exposing tokens to client-side scripts. To protect sensitive pages like /groups and /dashboard, Clerk's middleware intercepts each request, checks the session's validity, and either grants entry for active sessions or reroutes invalid or expired sessions to /auth/login. This end-to-end workflow combines flexibility, user convenience, and robust security across all authentication scenarios

Security Implementation

Clerk secures every step of your authentication workflow with built-in protections and configurable policies. For CSRF protection, Clerk automatically generates and validates per-session CSRF tokens for all form submissions whether signing up, logging in, or enrolling in a course ensuring that only forms served by your own domain can successfully submit sensitive requests. To thwart brute-force attacks, Clerk applies rate-limiting on its authentication endpoints: by default, only five login attempts are permitted per IP address per minute, and you

can tighten or loosen this throttle in the Clerk dashboard. Once a user is authenticated, Clerk issues a JWT that not only carries standard session data (user ID, expiration) but also embeds custom role claims for example, user.role === "instructor" or "admin" so that your middleware can enforce fine-grained access controls in admin panels or instructor-only routes. All JWTs are stored in HTTP-only, secure cookies and automatically refreshed just before expiry, giving you a turnkey solution that balances developer convenience with enterprise-grade security

User Management

After users sign in, Clerk's built-in User Profile component lets them easily upload avatars, write personal bios, and adjust display settings all stored as part of their Clerk identity and surfaced throughout the app to foster a sense of community. Behind the scenes, Clerk tags each session with custom role claims (for example, user.role = "instructor"), which your backend then enforces using Supabase Row-Level Security (RLS) policies. When an instructor-only operation like creating or editing a course is requested, your middleware reads the JWT's role claim before allowing the action, ensuring that only properly authorized users can access sensitive administrative endpoints. This tight integration between Clerk's profile UI and Supabase's RLS gives you end-to-end user management with minimal custom code

Integration of Clerk with the LMS Workflow
The integration of Clerk into our LMS ensures secure, role-aware access control while enabling seamless collaboration and real-time learning. The following sections map Clerk's functionality onto our workflow, illustrating how authentication, session management, user profiles, group-based learning, course operations, and security measures all work together.

Authentication & Session Management

Workflow Step: Authentication Method → Sign In/Up → Session Creation

- **Multi-Provider Authentication**
    - **Email/Password**
      Users submit credentials via a secure form. Clerk validates these against our Supabase user database and sends a confirmation link to verify email ownership before activating the account.
    - **Google OAuth**
      Users click "Continue with Google," complete Google's consent screen, and are redirected back to Clerk for token validation and profile synchronization.
- **Session Creation & Validation**
  Clerk issues JWT-based sessions stored in secure, HTTP-only cookies. Sessions auto-refresh every 15 minutes to prevent unauthorized access.

Protected Route Access

Workflow Step: Middleware → Route Protection → Valid Session?

- **Route Protection**
  Protected routes such as /group/[id] or /dashboard require an active Clerk session.
- **Role-Based Access Control**
  Session JWTs carry custom claims (e.g., user.role === "instructor") that your server code and

User Management & Profile Customization

Workflow Step: User Profile Management

- **Database Synchronization**
  After authentication, Clerk triggers webhooks that create or update user records in Supabase, linking the clerkId to LMS-specific roles:
- **Profile Customization**
  Users update avatars, bios, and display settings via Clerk's built-in UserProfile component. These settings are stored in Clerk and shared across the LMS to foster community engagement.

Group-Based Learning Structure

Workflow Step: Group Routes

- **Dynamic Group Access**
  Enrollment status determines group membership. Supabase RLS policies enforce access:
- **Real-Time Collaboration**
  Supabase Realtime powers live chat, posts, reactions, and presence tracking for active users.

Course Management & Security

Workflow Step: Course Creation → Content Management

- **Secure Course Operations**
  Instructors access /course/create only with a valid Clerk session. File uploads (PDFs, videos) use signed URLs to guarantee authorized access.
- **Privacy Controls**
  Courses can be marked public or private; membership-based access is enforced by Clerk-authenticated group assignments.

Real-Time Features & Analytics

Workflow Step: Real-Time Learning → Progress Tracking

- **Live Updates**
  WebSocket connections (via Supabase Realtime) sync course progress, chat messages, and notifications across devices.

- **Progress Tracking**
  User activities (e.g., completed modules) are recorded in Supabase and exposed only to valid Clerk sessions.

Security & Error Handling

Workflow Step: Security Implementation

- **CSRF Protection**
  Clerk embeds per-session CSRF tokens in all forms to block cross-site request forgery.
- **Rate Limiting**
  Login endpoints throttle attempts to five per minute to mitigate brute-force attacks.
- **Error Recovery**
  Expired sessions automatically redirect users to login, preserving unsaved progress in local cache.

Key Benefits of Integration

1. **Security**
    - End-to-end encryption for private messages.
    - GDPR-compliant session handling.
2. **Scalability**
    - Supports 10,000 + concurrent users leveraging Next.js and Supabase.
3. **User Experience**
    - Instant authentication via Google OAuth.
    - Mobile-first and dark-mode designs for maximum accessibility.

4.4 Database Management with Supabase

Supabase is an ideal choice for this LMS because it combines the reliability of a relational database with built-in real-time capabilities. Its PostgreSQL foundation ensures strong data integrity, complex querying, and enforceable schemas critical for storing course records, user profiles, and permissions. At the same time, Supabase's Realtime API and on-database triggers push updates instantly to connected clients, powering live chats, progress tracking, and collaborative features without additional infrastructure. This hybrid model accelerates development by providing authentication, storage, and real-time subscriptions out of the box, while preserving the transactional guarantees and ACID compliance required for secure, scalable operations.

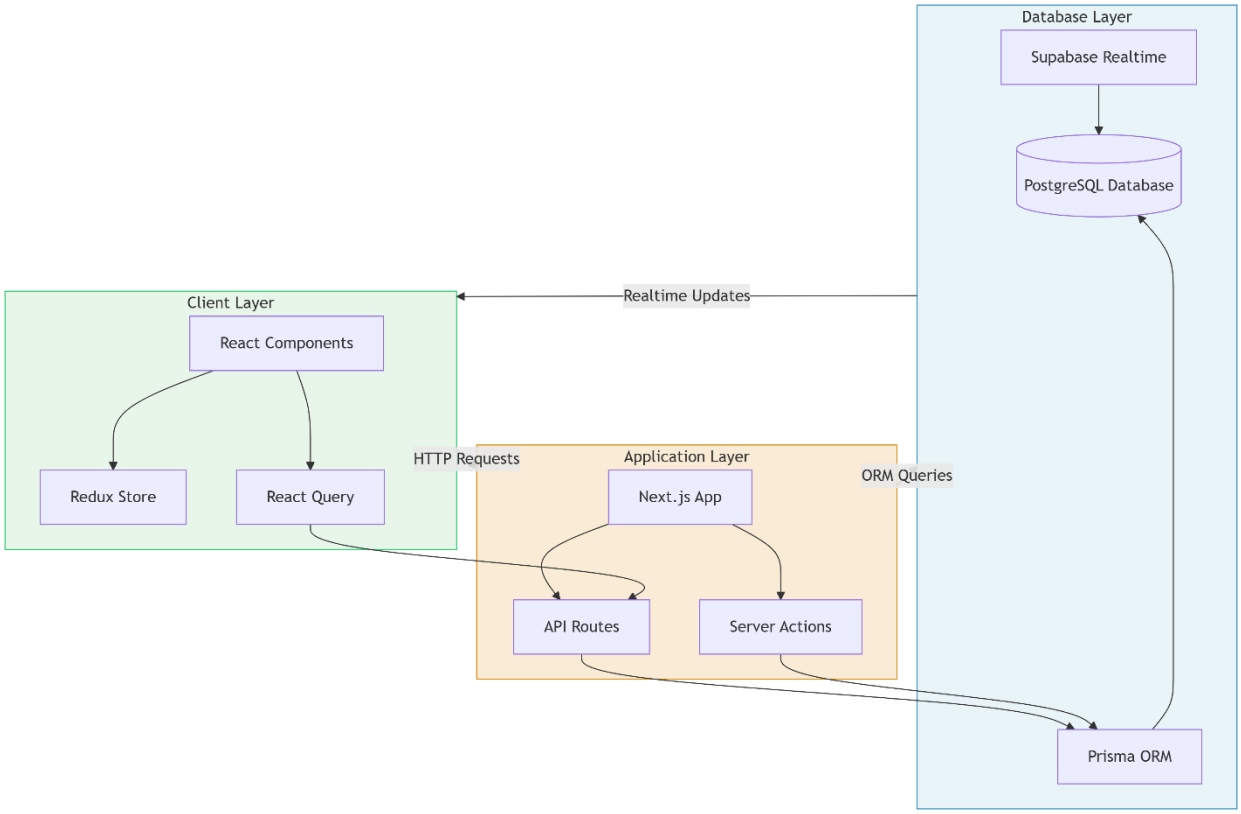


Figure 3 Database Relation

The database architecture combines Prisma ORM for structured relational management and Supabase Realtime for dynamic interactivity, creating a robust foundation for the LMS's social and pedagogical workflows. Below is a detailed breakdown of the system's data layer, aligned with the provided schema and architecture diagrams

Hybrid Architecture

The system employs a layered architecture to balance efficiency and real-time responsiveness:

Client Layer

- **React Components**: Fetch data via React Query (e.g., useQuery('posts', fetchGroupPosts)).
- **Redux Store**: Manages global state (e.g., authenticated user's group memberships).
- **Offline Support**: React Query caches data locally, allowing access to recent posts/courses without connectivity.

Application Layer

- **Next.js API Routes**: Securely interact with the database (e.g., POST /api/groups creates a new group).

- **Prisma ORM**: Executes typed queries (e.g., prisma.user.findMany({ where: { groupId } })).
- **Server Actions**: Handle transactional logic (e.g., enrolling users in courses).

Database Layer

- **PostgreSQL**: Stores structured data (users, groups, posts) with ACID compliance.
- **Supabase Realtime**: Powers live features via WebSocket subscriptions:

Database Schema Design with Prisma

The PostgreSQL schema, defined via Prisma ORM, enforces type safety and relational integrity across core entities

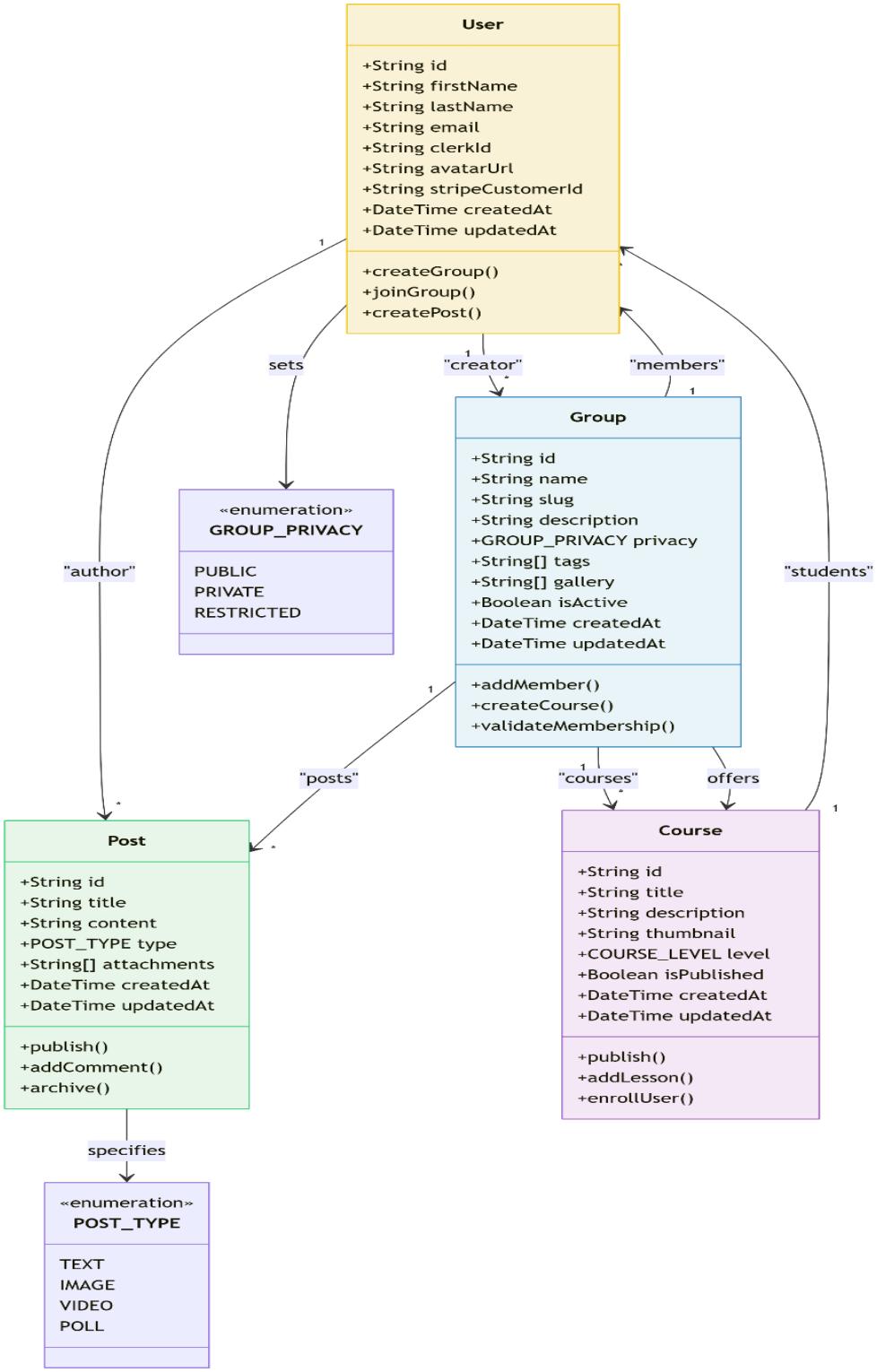


Figure 4 Database Schema

we define a robust PostgreSQL schema using Prisma ORM to guarantee type safety and maintain relational integrity across all core entities. For example, the User model identified by a UUID and linked to Clerk via a unique clerkId relates to both groups (as a member or creator) and posts (as an author), ensuring that user-driven content and memberships are consistently enforced. The Group model includes fields like name, privacy (an enum with PUBLIC, PRIVATE, or RESTRICTED values), and an explicit creator relationship back to User, while its arrays of posts, courses, and members capture the full scope of collaborative activity. Similarly, the Post model ties titles, content, and attachments to both User and Group, and the Course model links lessons and enrolled users to its parent group, with an enum-defined COURSE_LEVEL to standardize difficulty. By codifying these relationships and constraints at the schema level, Prisma helps catch mismatches at compile time, automatically generates type-safe database clients, and upholds ACID guarantees providing a solid foundation for both complex queries and real-time updates via Supabase.

Security & Performance

security and performance are enforced at the database level using Supabase's Row-Level Security (RLS) and Prisma's query optimizations. For example, an RLS policy on the posts table CREATE POLICY "Group Post Access" ON posts USING (group_id IN (SELECT group_id FROM members WHERE user_id = auth.uid())); ensures that only authenticated members of a given group can read its posts. On the performance side, we leverage Prisma's include and select options to fetch exactly the data needed (for instance, pulling only post titles when rendering a feed), reducing payload sizes and network overhead. Finally, by indexing primary lookup fields such as groupId and clerkId, we guarantee sub-100 ms query responses even under heavy load. This combination of fine-grained access control and carefully tuned queries provides both robust security and the high throughput required for a responsive, real-time learning experience.

4.5 Application Interface Flow

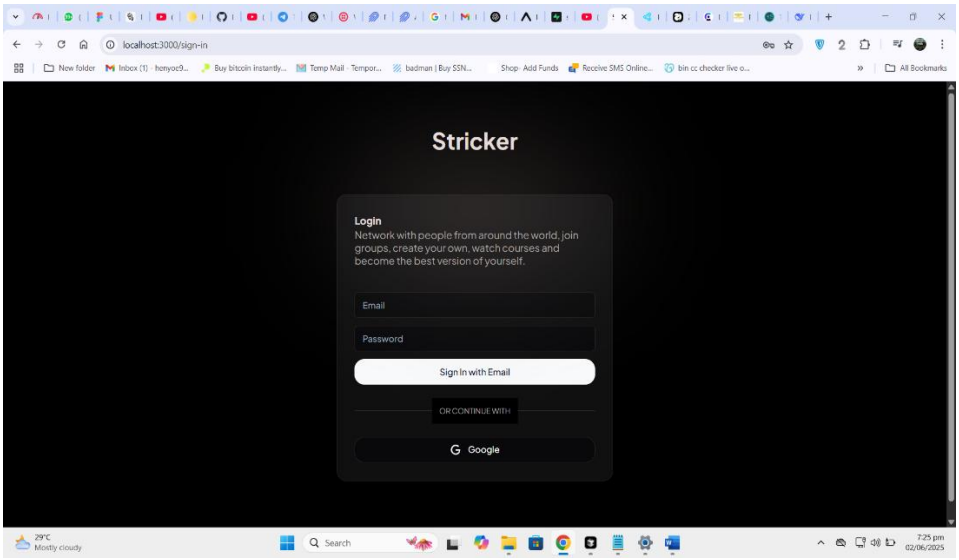


Figure 6 Login Page

User Register or Login if they already have an account , if Auth go User is been direct to Explore Page where they see available course on the platform

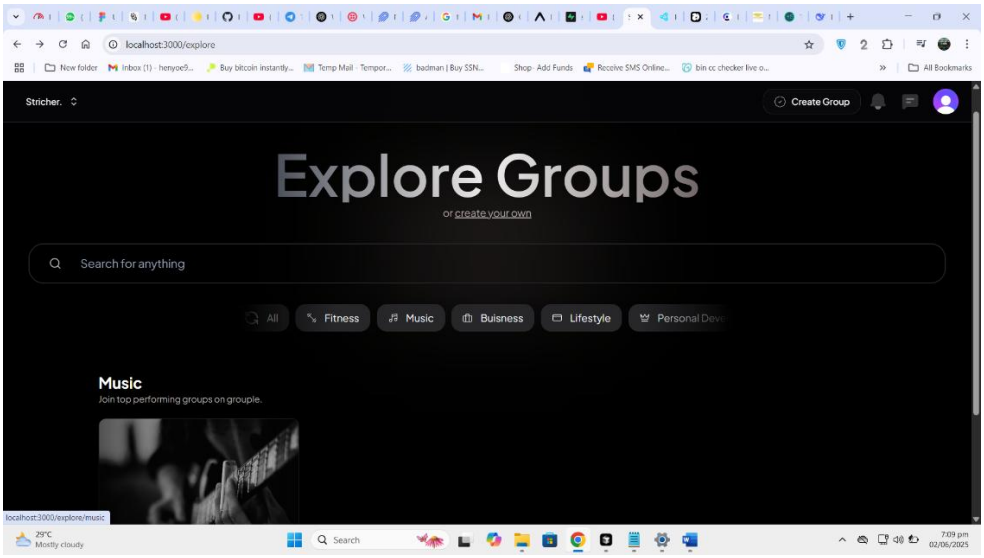


Figure 7 Explore Page

If User cant find the particular Group they are looking you can use the search feature to get the group e.g below

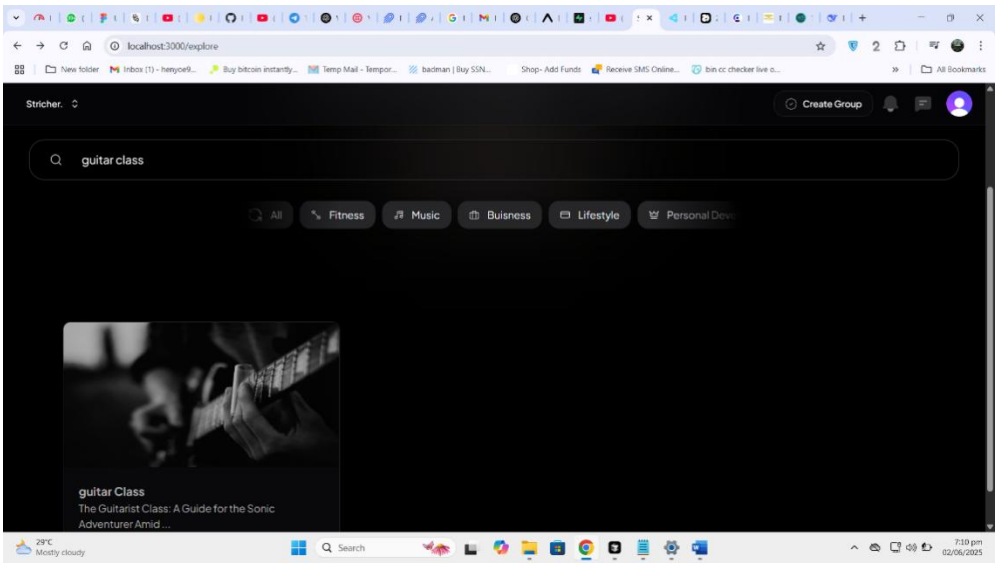


Figure 7 Explore Page (FILTERED)

If User want to create their own group then click on the create Group Botton at the top , where the user profile ,To Create Group

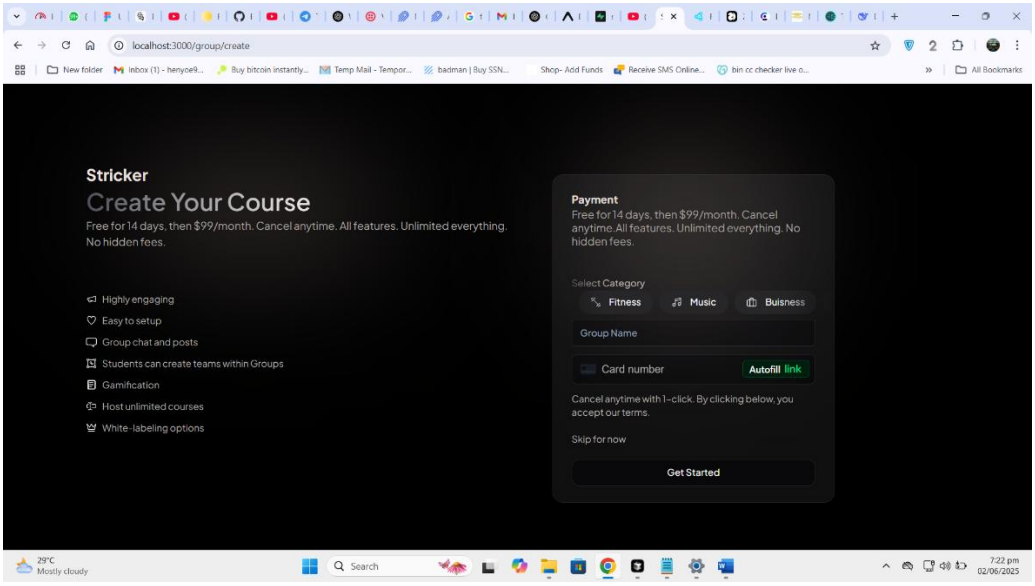


Figure 9 Create Course Page

User will fill the necessary form and click on get started to create a group and you will redirected to the Group interface

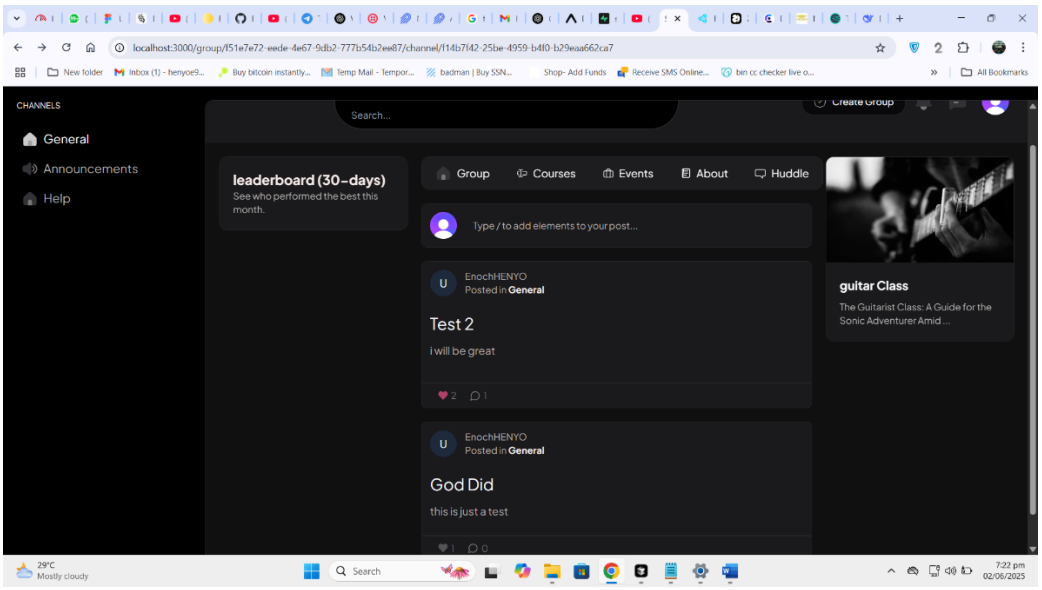


Figure 10 Dashboard

User can add Create Material for student to learn from by Clicking on the Course Tap , if only you are the Group owner , you are given the Access to create new or add material

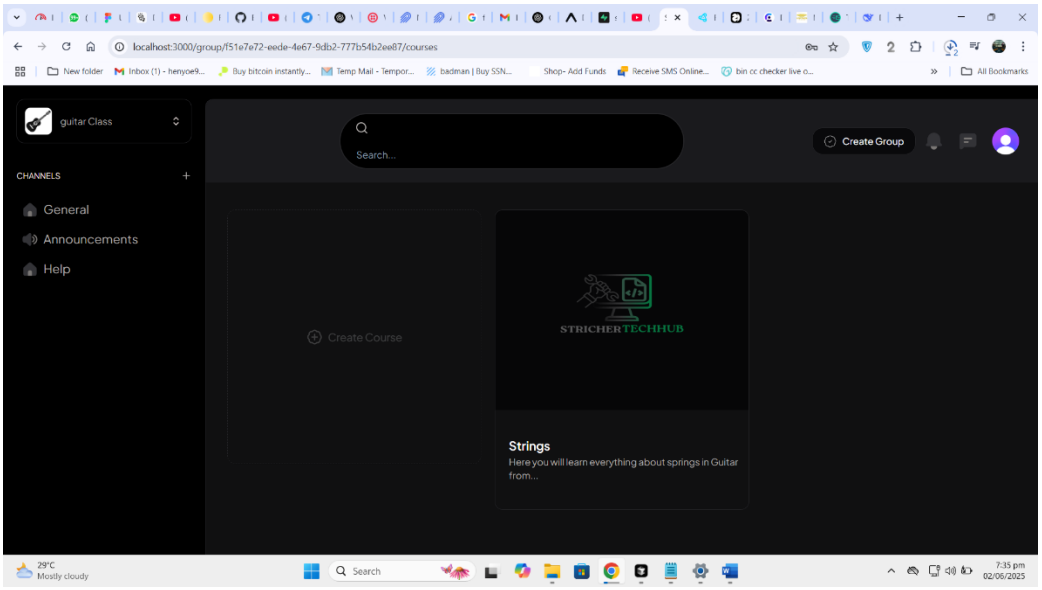


Figure 11 Add course Page

Messaging

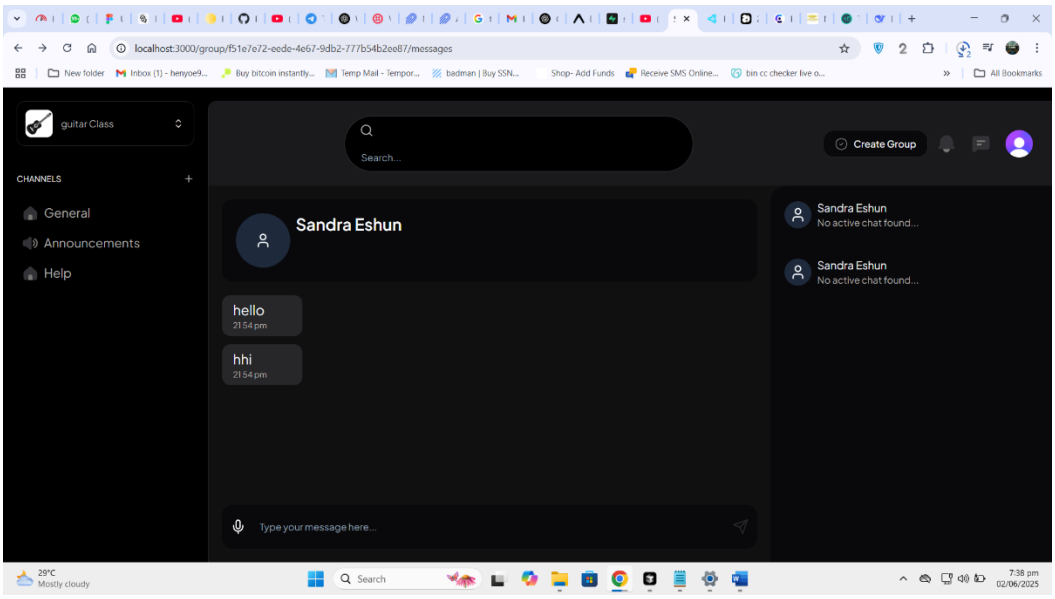


Figure 12 Messaging Page

When a student Register or Login and want to enroll in your Group or Course , He or she will see the payment , depending on how much course owner or Group is selling it ,in the figure below , a Student want to enroll in Group or Course that cost 10$

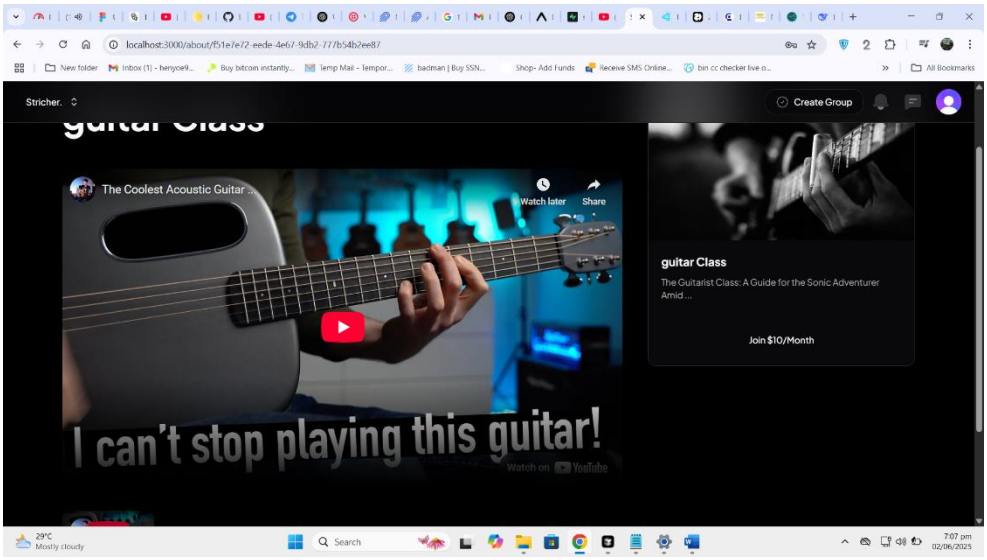


Figure 13 : Group Landing Page

Chapter 5

SUMMARY, LIMITATION AND CONCLUSION

5.1 Introduction

This chapter provides a summary of the key findings from the study, outlines practical recommendations based on the results, and presents the final conclusion. It reflects on the goals of the project, evaluates how they were achieved, and suggests ways to improve future implementations of the proposed Saas Based Learning Management System

5.2 Summary Of Findings

This study aimed to design a modern Learning Management System (LMS) that meets the specific needs of Central University, Ghana. The research began by examining existing LMS platforms and identifying several key problems. These included low student engagement, limited collaboration tools, poor mobile performance, and difficulty scaling to support more users. To address these issues, a new LMS was proposed that uses modern web technologies like Next.js for fast and interactive interfaces, Clerk for secure user authentication, and Supabase for real-time data and communication. The new system focuses on being user-friendly, engaging, and accessible for all types of learners. The study was also guided by strong educational theories. Learning principles such as social constructivism and connectivism highlight the importance of collaboration, interaction, and networked learning. These theories support the idea that students learn better when they can share ideas and work together in real time. Many traditional LMS platforms fail to support these needs, often relying on outdated tools like static discussion forums. The proposed LMS improves on this by offering features like activity feeds, group chats, and personalized dashboards encouraging real-time engagement and peer-to-peer learning. To ensure the new system was practical and achievable, a detailed analysis of current LMS use at Central University was conducted. The findings confirmed that the existing platforms were not meeting the needs of students or teachers. As a result, a flexible, cloud-based system was designed with features that encourage communication, collaboration, and easy access on all devices. A feasibility study showed that the project was both technically and financially realistic, with cloud technology providing the needed scalability and performance. The development process follows an Agile approach, which allows the team to build the system step by step, testing and improving it based on user feedback. The plan includes collecting requirements from students and teachers, designing and building the platform, testing it in real-life situations, and continually improving it based on how people use it. This ensures the final product is closely aligned with the needs of its users.

In the development phase, the system was built using modern tools and carefully organized into clear sections, such as login pages, course views, and group workspaces. Security, performance, and user experience were key priorities. Real-time updates and easy-to-use interfaces ensure that students and teachers can interact without delays or confusion. The system is also mobile-friendly and works well even in low-connectivity areas, making it accessible to a wider range of users.

In conclusion, the research shows that with the right design and technology, a Learning Management System can become much more than just a tool for uploading content. It can be a space where students engage, collaborate, and grow. This new system sets a higher standard for digital learning in African universities by combining strong learning theory, technical excellence, and real user needs.

5.3 Limitation

One limitation of the web application is its video hosting solution. Instead of storing videos directly within the system, course videos are hosted externally on platforms like Loom or YouTube, with only the video links embedded in the app. This approach presents several potential drawbacks: it restricts offline access to video content, makes the application dependent on the reliability and performance of these third-party services, and means video availability is subject to the hosting platforms' terms and conditions. While this solution helps manage storage requirements, it does create some dependence on external services that could affect the user experience.

5.4 Futures Works

To further enhance the platform’s impact on digital learning, several features are planned for future development. First, an Event Tab will be introduced to serve as a central hub where all upcoming group activities, academic events, and announcements can be posted. This will improve communication and ensure students remain informed and involved in both academic and social aspects of their learning environment.

In addition, a Huddle Tab is proposed to create a space for real-time interaction and informal collaboration. This feature will support peer-to-peer engagement, encourage group discussions, and provide a virtual “gathering spot” that mimics in-person study groups, ultimately building a stronger sense of community among learners.

To encourage continuous participation and academic excellence, a Leaderboard feature will be implemented. By tracking student activity, contributions, or performance, this gamified element aims to promote motivation, friendly competition, and consistent engagement across the platform.

A dedicated Assignment Feature will also be developed to allow tutors to create, distribute, and assess assignments directly within the platform. This will reduce reliance on external tools and ensure a more streamlined workflow for both students and instructors.

Finally, significant improvements will be made to the Course Page Interface to enhance usability and functionality. The goal is to provide a more seamless experience for tutors uploading course materials and for students navigating and consuming learning content. A cleaner design, better organization, and mobile responsiveness will ensure accessibility and ease of use across devices.

These planned enhancements are designed to make the application not only a robust academic tool but also a socially engaging and user-centered platform for 21st-century learning

5.5 Conclusion

In conclusion, this SaaS-based LMS represents a significant evolution in digital learning platforms, successfully bridging the gap between academic rigor and modern digital engagement. By leveraging cutting-edge technologies like Next.js for responsive interfaces, Clerk for secure authentication, and Supabase for real-time data management, we've created a system that addresses the critical limitations of traditional LMS platforms while introducing innovative social learning features. The platform's cloud-native architecture ensures scalability to accommodate Central University's growing needs, while its intuitive design and mobile-first approach guarantee accessibility for all users. Although certain technical constraints like external video hosting exist, the overall solution delivers a robust, future-proof foundation for transformative education. This LMS not only meets current pedagogical requirements but also establishes a framework for continuous innovation in digital learning, setting a new standard for educational technology in African universities and beyond. Its successful implementation demonstrates how thoughtful technology integration can enhance learning outcomes while fostering collaborative, engaging educational experiences.

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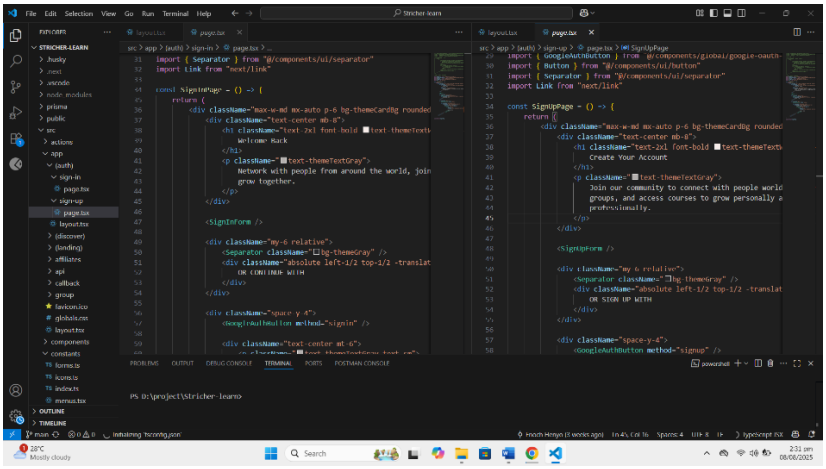
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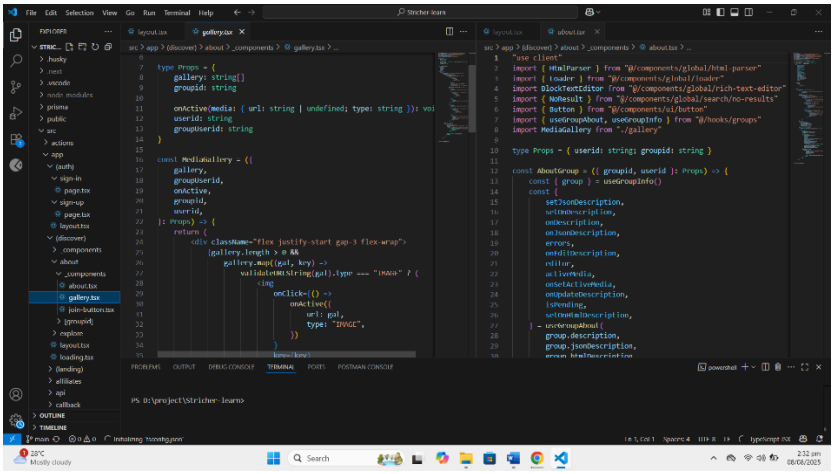
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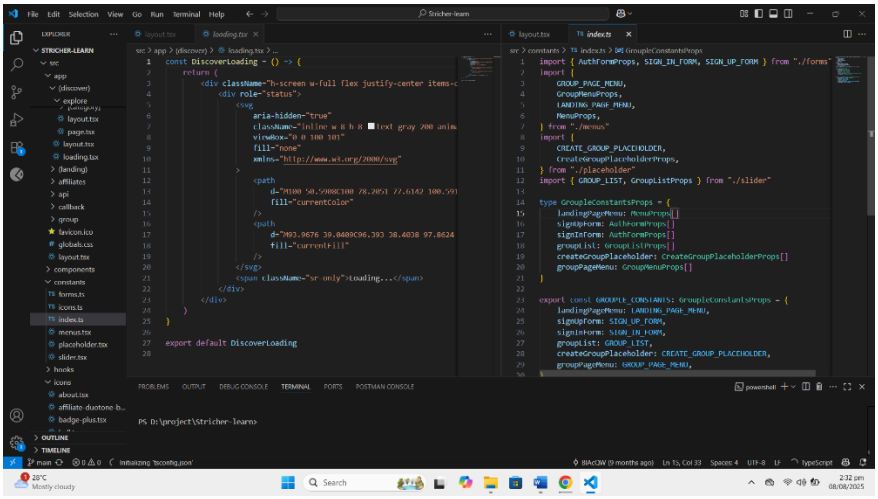
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Appendix 1

Some codes







Appendix 2

Setup Instructions for my Project

1. Clone the Repository

git clone https://github.com/BlAcQW/Stricher-learn.git

cd Stricher-learn

2. Install Dependencies

Make sure you have Node.js (v18+) and npm or yarn installed. If now go to https://nodejs.org/en/download download and install Node.js

Then run:

npm install

3. Create a .env File

In the project root, create a .env file and add the following variables**:**

DATABASE_URL=

NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY=

CLERK_SECRET_KEY=

NEXT_PUBLIC_STRIPE_PUBLISH_KEY=

STRIPE_SECRET_KEY=

UPLOAD_CARE_PUBLIC_KEY=

UPLOAD_CARE_SECRET=

PROJECT_ID_VERCEL=

TEAM_ID_VERCEL=

AUTH_BEARER_TOKEN=

NEXT_PUBLIC_SUPABASE_ANON_KEY=

NEXT_PUBLIC_SUPABASE_URL=

4. Explanation of Environment Variables

- DATABASE_URL → The connection string for your database (e.g., PostgreSQL).
- NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY → Public key for Clerk authentication (frontend).
- CLERK_SECRET_KEY → Secret key for Clerk authentication (backend).
- NEXT_PUBLIC_STRIPE_PUBLISH_KEY → Public key for Stripe payments (frontend).
- STRIPE_SECRET_KEY → Secret key for Stripe payments (backend).
- UPLOAD_CARE_PUBLIC_KEY → Public key for UploadCare file uploads.
- UPLOAD_CARE_SECRET → Secret key for UploadCare authentication.
- PROJECT_ID_VERCEL → Your Vercel project ID (for deployment).
- TEAM_ID_VERCEL → Your Vercel team ID (for deployment).
- AUTH_BEARER_TOKEN → Authentication token for Vercel API.
- NEXT_PUBLIC_SUPABASE_ANON_KEY → Public anonymous key for Supabase (frontend).
- NEXT_PUBLIC_SUPABASE_URL → Supabase project URL.

5. Run the Development Server

npm run dev

Then open in your browser

http://localhost:3000

Appendix 3

Similarity Report (TurnitIN)